THE NORM OF AN AMPLE INVERTIBLE SHEAF ALONG A FINITE MORPHISM WITH A DEGREE D NORM IS AMPLE ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 3737-3742 and proof span 3744-3770 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof names the norm sheaf and uses the author's notation for the norm of a section; both are preserved unchanged.

Editorial note. The proof's conclusion uses the cap criterion with the phrase that the nonvanishing locus is affine; the author's wording is kept.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A scheme carrying an ample invertible sheaf is quasi-compact, and the existence of a norm of degree d implies that the morphism is surjective. Cited at proof lines 3746-3749. Gives quasi-compactness of the base, which is the first half of the ampleness criterion being applied.
- Ampleness tested by affine caps: it suffices to find, at every point, a section of a positive tensor power not vanishing at that point whose nonvanishing locus is affine, and the existence of a section of a high power of an ample sheaf whose nonvanishing locus separates a finite set from a prescribed affine open. Cited at proof lines 3752-3761. Produces the section downstairs whose nonvanishing locus lies inside a chosen affine open and whose behaviour at the norm is controlled.
- The norm of a section: for a finite morphism with a degree d norm the norm of a section of a power of an invertible sheaf is a section of the same power of the norm, whose vanishing behaviour is controlled by the section's vanishing behaviour. Cited at proof lines 3762-3764. Identifies the candidate sheaf on the base as ample and gives the nonvanishing statement.

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- The nonvanishing locus of a section of a sheaf restricted to an affine open is affine. Cited at proof lines 3765-3767. Concludes that the cap of the chosen section is affine, completing the ampleness criterion.

Source reference. Original lemma 0BCZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14297–14297. Same-chapter reference used in the retained author text at source line(s) 3768. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 01PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2241–2241. Same-chapter reference used in the retained author text at source line(s) 3751. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01PV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2244–2244. Same-chapter reference used in the retained author text at source line(s) 3769. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 09NV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12228–12228. Same-chapter reference used in the retained author text at source line(s) 3254, 3765. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes. Assume X has an ample invertible sheaf and there exists a norm of degree d for π. Then Y has an ample invertible sheaf.*

Proof. Let $\mathcal{L}$ be the ample invertible sheaf on X given to us by assumption. We will prove that $\mathcal{N} = \mathrm{Norm}_\pi(\mathcal{L})$ is ample on Y.

Since X is quasi-compact (Properties, Definition 01PS) and $X \to Y$ surjective (by the existence of Norm_π) we see that Y is quasi-compact. Let $y \in Y$ be a point. To finish the proof we will show that there exists a section t of some positive tensor power of $\mathcal{N}$ which does not vanish at y such that Y_t is affine. To do this, choose an affine open neighbourhood $V \subset Y$ of y. Choose $n \gg 0$ and a section $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that

$$\pi^{-1}(\{y\}) \subset X_s \subset \pi^{-1}V$$

by Properties, Lemma 09NV. Then $t = \mathrm{Norm}_\pi(s)$ is a section of $\mathcal{N}^{\otimes n}$ which does not vanish at x and with $Y_t \subset V$, see Lemma 0BCZ. Then Y_t is affine by Properties, Lemma 01PV. □